Tulsi Leaf Disease Detection using CNN

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Abstract - Since the start of the covid 19 pandemic, a wide range of medications have been produced and are currently being utilized to treat the disease. Tulsi, in addition to all of the chemical-based medications, is an herbal therapy that is particularly effective in the treatment of this ailment. Tulsi has been used to heal ailments and infections for millennia, particularly in India. Because we use tulsi for medicinal purposes, it's vital to monitor its health in order to reap the full benefits of its herbal properties. Plant diseases harm the health and growth of the plant. Disease detection in plants is crucial so that it can be treated before it spreads throughout the plant. To detect illnesses in tulsi leaves, we propose employing a model based on convolution neural networks. Image processing and CNN are widely employed. The prepared model extracts the image's key features and categorizes it into different disorders. The model has a 75 percent accuracy rate.

Keywords: CNN, Tulsi leaf, Disease Detection, Image's key features.

I. INTRODUCTION

Plants are an essential aspects of human life cycle. Medical plants are plants that are utilized for their unique qualities that benefit human health. They were first referred to as simple in medieval medicine, and they now refer to medicinal plant products from traditional or modern herbal therapy [1]. Detection of plant disease today require some advance instruments and technology, but this facility is not available everywhere. To check infected plants, one need to carry the samples to a nearby lab or facility to get it tested. This process is inefficient and time consuming. Infected crops should be detected before it affects every other crop in the field.

The photos were filtered using the Gabor filter in 2012, and the Artificial Neural Network (ANN) classifier was utilized. The retrieved feature values are then trained with a classifier that has a 91% accuracy rate. Deep convolutional neural networks (CNN) are used in agriculture to recognize and classify plant diseases. For forecasting tomato plant disease, a deep learning Convolutional Neural Network architecture trained with machine learning is presented.
To improve precision and accuracy, another model is developed that combines segmentation and CNN with an 8 convolution layer model. The paucity of training data, lack of creativity in the method, and insufficient processing power hampered CNN's performance in complicated tasks [2].

A learning system that is based on transfer-learning CNN is used on a MobileNetV2 and an EfficientNetB0 model. Each model's layer weight was frozen before the totally linked layer, and all layers were then removed [2]. Since the beginning of the Covid 19 epidemic, a lot of research has been done to discover a treatment. Tulsi, in addition to all other medications, has been shown to be a successful treatment for covid symptoms. Tulsi is a natural plant with several health advantages and no known adverse effects. Tulsi can help to treat throat infections, balance kapha and vata, and boost immunity [3]. However, the infectious tulsi leaves can be hazardous to the treatment of other diseases since they can cause harmful bacteria to proliferate inside the body. Infected leaves can also harm the entire plant, lowering its medicinal benefits. Our study identifies pathogens in plants so that preventative actions can be implemented.

A. *Tulsi Leaf*

Tulsi is a medicinal plant that has been utilized for thousands of years in sectors such as Ayurveda, medical research, and more. Tulsi is known for its healing powers and has a comprehensive significance in Indian culture. The leaves are mainly pointy and oval in shape, and can be up to 2-6 cm long [4].

Fig: 1 Tulsi Plant

B. *Types of Tulsi Leaf Disease*

a. *Fusarium Wilt*

Fusarium wilt is a fungal disease that can be transmitted from the soil or the seeds of the infected basil plants. The basil infected by this disease are wilted and yellow.

Fig: 2 Tulsi with Fusarium Wilt infection

b. *Powdery Mildew*

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Powdery mildew is also a fungal infection which is caused by parasites. Powdery mildew fungi belong to order Erysiphales are most common obligate parasites [5]. On the host surface, these fungi produce enormous conidia, which leave a white powdery covering and a dusty look.

Fig: 3 Tulsi with Powdery Mildew infection

c. Tulsi shoot blight

The shoot blight is caused by C. gloeosporioides, a fungal disease of tulsi. Tulsi leaf have circular spots which later turns into pinkish spots. Tulsi leaves were observed with circular, white colored spots, which later coalesced and resulted in necrosis of leaves causing severe reduction in yield [6].

Fig: 4 Tulsi with Shoot Blight Infection

II. METHODOLOGY

Using traditional machine learning approaches, achieving the classification problem takes numerous sequential phases, including pre-processing and feature extraction [7]. For picture categorization, the project's technique employs Neural Networks. Unsupervised machine learning techniques such as neural network (NN) classify unlabeled data [8].

A. Neural Networks

In the autonomous identification of illnesses in plant leaves, neural network (NN) are applied. It is well-known method because it has as an effective classifier for many real-world problems, neural networks (NN) were chosen as the instrument to classify photos [9]. Neural networks (NN) contains variety of shapes and sizes, depending on the application. Recurrent neural networks and convolutional neural networks are two types of networks that are extensively employed. We employed a convolutional network in this research (CNN).

B. Convolution Neural Network

A CNN is a form of machine learning method that works with input that has a grid layout, such digital photos. Pixels are placed in grids in the image structure, and each pixel has a unique property. CNN draws its name from "convolution matrices," a mathematical linear process between matrices [10]. In the earliest layers of the CNN algorithm, basic characteristics such as lines and curves are detected, while more complex features (patterns and faces) are detected in advance levels.

A CNN typically has three layers:

- Convolutional layer
- Pooling layer
- Fully connected layer [11].

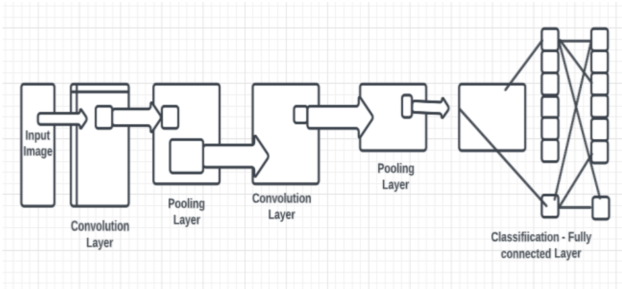

Fig: 5 Proposed CNN Model

The model of the system takes a database of leaf images. Then the data is cleaned and classified into two parts: Training and Testing Dataset

The training dataset is now pre-processed for further implementations.

Pre-processing of images include:

- Setting the width and height of the images.
- Setting the color orientation.
- Setting the input size of the images.

After the pre-processing the image will go to the CNN layers and it will first get classified into categories: Infected and non-infected. Finally, the images categorized as infected will be differentiate further for prediction of different types of diseases[12].

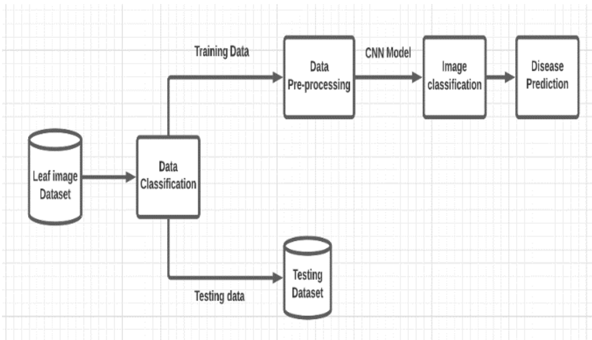

Fig: 6 Process Flow Diagram

C. Image segm entation in CNN:

The goal of the image segmentation is to segment the plant leaf digital image into various parts and partitioning is called segmentation. It's a tool for detecting image features Picture segmentation recognizes a variety of features in a picture, including Line, Point, and Boundary of specific areas [13]. It assists with the recognition of distinguishing aspects in a photograph [14].

D. Feature Extraction:

We do feature extraction after segmentation. It's a method for determining unique elements that represent a picture by taking into account the color and texture of the image[15]. To test the model's accuracy, the testing dataset will be introduced into the trained CNN model.

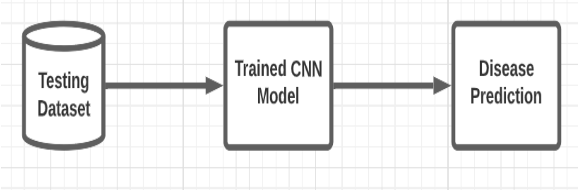

Fig: 7 Data Flow Diagram

E. Use Case Diagram:

The users of our system will upload images of plant leaves to the server after which the images will be processed and classified in categories i.e. infected and non-infected and finally the infected leaves disease will be predicted. The predicted output is sent back to the user.

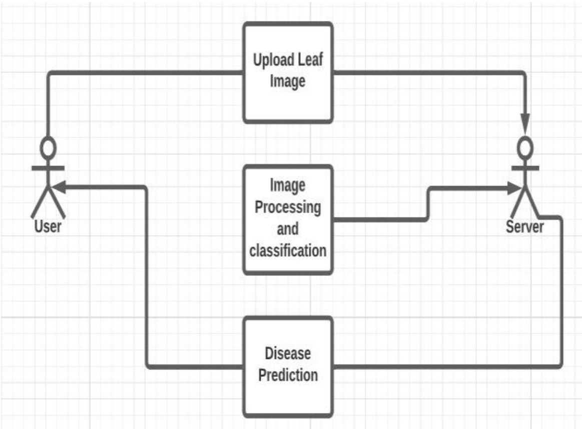

Fig: 8 Use Case Diagram

III. Results and Discussion

The main challenge we faced during the development was the availability of the Tulsi Leaf database. As there was no dataset available on internet for Tulsi leaf, we have to create our own database by collecting tulsi images of different varieties. Another challenge we faced was to design a model to recognize different patterns in various diseases of tulsi images because tulsi leaves are small in size as compared to other leaves. Due to the small size of leaf it was difficult to detect similar pattern with accuracy [16].

To achieve good accuracy from our training model, we needed a large dataset. But due to unavailability of large database for tulsi images we used image augmentation. Image augmentation is used to generate training images by performing multiple processes like rotation, shifting, flips, etc. Image generated from augmentation can be used in training of the machine learning model. As a result, 60 new tulsi dataset pictures were produced using Image Augmentation, on which the model was trained [17-18].

Fig: 9: *Augmented Images gathered from a single image*

Precision is the measure of percentage of tuples classified as positive are actually positive, whereas recall depicts percentage of positive tuples are labeled as such.

- *Precision = TP / TP+FP*
- *Recall = TP / TP+FN*

The average accuracy of the model is depicted by the table 1. The table depicts the average of two datasets tested on the model.

Table 1: Table depicts precision, recall and F1- score

	precision	recall	Fl-score	support
0	0.78	0.75	0.74	136
1	0.74	0.75	0.72	130
avg/total	0.76	0.75	0.73	266

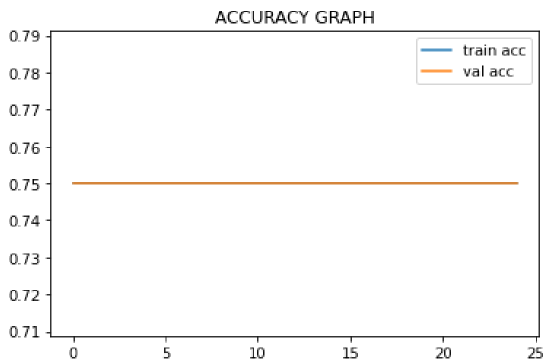

Fig: 10: Accuracy Curve

Figure 10 (Accuracy curve) depicts the accuracy of the model over different data epochs. Figure 11 (Loss curve) illustrates the loss percentage over different data epochs.

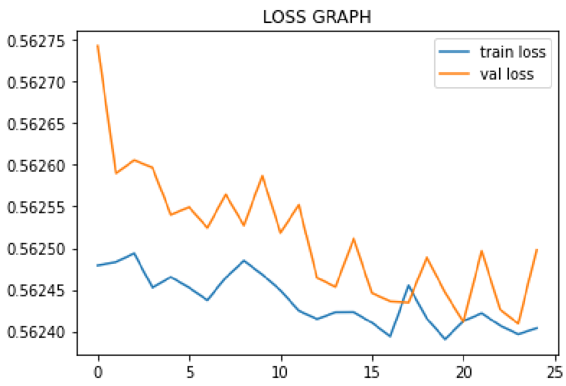

Fig: 11: Loss Curve

IV. CONCLUSIONS

Tulsi has a wide range of health benefits. Tulsi has been shown to be beneficial in a variety of situations, including immunity and cardiac problems. In medicinal plant research, disease detection in Tulsi leaves could be crucial. With the increased use of tulsi leaves in the medical area, a technique to identify diseases in leaves and prevent further contamination is needed, so that those in need might receive high-quality herbal medicine. To analyze photos and detect infections, our suggested approach is to use a CNN model. In the discipline of image processing, CNN is the most widely proposed model. It is frequently employed to investigate visual imagery. We used CNN in our project since previously created models for image processing utilizing CNN were demonstrated to be accurate. We used picture augmentation to enhance the sample size of the data because there was no database for diseased tulsi photographs. Our model's results indicated a 75% accuracy rate, and it was able to correctly forecast illnesses in the majority of the photos.

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